

Rik Kumar

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EDUCATION

University of Waterloo

Bachelor of Computer Science, Honours, Co-op

2024 – Expected 2029

GPA: 3.8

EXPERIENCE

City of Edmonton

Applied AI Intern

Jan. 2026 - Aug. 2026

Edmonton, AB

- Refactored a dual-container architecture into a **Python/Flask thin-client system**, reducing dashboard load times by over **65%** through optimized server-side spatial joins and a streamlined **React/Leaflet** interface for real-time neighborhood analytics.
- Engineered a **GluonTS DeepAR** forecasting model using a **Negative Binomial** output layer to handle sparse, low-volume incident data, achieving **MAE of 0.02** and **RMSE of 0.1231** through adaptive scaling.
- Developed an **XGBoost-based predictive pipeline** for crime forecasting, implementing cyclic temporal encoding, multi-scale lag features, to achieve an **AUC-ROC of 0.88** on sparse incident data.
- Architected an asynchronous **batch-processing NLP** pipeline using Python and the Gemini API to classify high-volume incident logs into granular call types via a standardized domain taxonomy, resolving ambiguity downstream of Snorkel labeling.

IBM Research Labs

Undergraduate Research Extern

Oct. 2025 - Nov. 2025

Remote

- Built a Qiskit pipeline to validate multi-qubit density matrices via automated trace-squared ($\text{Tr}(\rho^2) = 1$) purity testing and eigenvalue analysis.
- Implemented automated error-handling routines to securely map validated pure states into state vectors for native quantum circuit initialization.

Estee Advisors

Tech Intern

May. 2025 - Aug. 2025

Hybrid

- Automated UI testing processes using **JavaScript**, **Node.js**, and **Selenium WebDriver**.
- Developed and optimized automated test scripts utilizing Excel.js for structured data processing, validation, and dynamic XPath management.

PROJECTS

LeetPush | Code | *Flask API, GraphQL API, Python, JavaScript, HTML*

Aug. 2025

- Developed a Chrome extension using **JavaScript**, **HTML**, and **GraphQL** to detect accepted LeetCode submissions and parse problem metadata.
- Built a **Python/Flask** backend to automate solution storage, code formatting, Git commits, and serve a local dashboard for tracking progress.

Winning FPL | Code | *Pandas, Numpy, Scikit-learn, Matplotlib*

Mar. 2025 – May. 2025

- Engineered a player-rating algorithm using FPL API data to optimize team selection, boosting overall rank by 300 % into the top 5th percentile.
- Built automated ETL pipelines using Pandas and NumPy to process historical player telemetry, fixture difficulty ratings (FDR), and rolling underlying metrics (xG/xA).

Commodity Price Forecasting | Code | *Pandas, Numpy, Scikit-learn, Matplotlib*

Dec. 2024 – Jan. 2025

- Built a SARIMA time-series pipeline to forecast commodity prices and model dynamic storage contract valuations using NumPy and Matplotlib.

Credit Risk Analysis | Code | *Pandas, Numpy, Scikit-learn*

Dec. 2024 - Jan. 2025

- Developed Decision Tree and Logistic Regression classifiers on FICO score data to predict loan default probabilities and automate risk-tier bucketing.

TECHNICAL SKILLS

Programming Languages: Python, Racket, C/C++, SQL, JavaScript

AI/ML: XGBoost, GluonTS (DeepAR), Scikit-learn, SARIMA/ARIMA

Frameworks and Libraries: Polars, Pandas, NumPy, Flask, Node.js, React, Leaflet, GraphQL, Selenium WebDriver, Qiskit